

Solution of differential equations

Linearisation of non-linear differential equations

Confine our attention to 2nd order ODEs

Consider

$$y'' + 3yy' + y^3 = 0$$

under appropriate constraints on domain, linear equations have two linearly independent solutions

$$Ly = 0 \quad L \text{ 2nd order linear operator}$$

$$\{y = y_1(x), y_2(x)\}$$

$$y'' + xy = 0$$

$$\begin{aligned} y_1(0) &= 1, & y_2(0) &= 0, \\ y_1'(0) &= 0, & y_2'(0) &= 1 \end{aligned}$$

$$y_1(x) \quad y_2(x)$$

$$y(x) = a_1 y_1(x) + a_2 y_2(x)$$

Two equations are equivalent if they possess the same Lie algebra of symmetries i.e. there exists a point transformation which transforms one equation to the other.

Point transformation

$$x, y \mapsto X, Y, \quad X = F(x, y); \quad Y = G(x, y)$$

other transforms depend upon the derivatives e.g. contact

$$x, y \mapsto X, Y, \quad X = F(x, y, y'); \quad Y = G(x, y, y')$$

Lie algebra

$\{G_i, \ i = 1, \dots, n\}$ forms a Lie algebra if

$$[G_i, G_j] = C_{ijk} G_k$$

where the C_{ijk} are constants (called structure constants) and $[,]$ is a skew symmetric operator

$$[G_i, G_j] = G_i G_j - G_j G_i$$

so that it is obvious that

$$[G_i, G_j] = -[G_j, G_i]$$

Note: the structure constants are invariant under a point transformation. i.e. the algebra is independent of the co-ordinate representation being used.

In general a symmetry exists if a "system" is invariant under the transformation induced by the symmetry. An infinitesimal point transformation of x, y to $\bar{x}, \bar{y}$ is represented by

$$\bar{x} = x + \varepsilon \xi(x, y), \quad \bar{y} = y + \varepsilon \eta(x, y),$$

where ε is the infinitesimal parameter. For $f(x, y)$ an infinitesimal point transformation produces

$$\begin{aligned} f(\bar{x}, \bar{y}) &= f(x + \varepsilon \xi, y + \varepsilon \eta) \\ &= f(x, y) + \varepsilon \left(\xi \frac{\partial f}{\partial x} + \eta \frac{\partial f}{\partial y} \right) + O(\varepsilon^2) \\ &= f(x, y) + \varepsilon Gf. \end{aligned}$$

where

$$G = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}$$

is the generator of the infinitesimal point transformation.

$$\left\{ G_i = \xi_i \frac{\partial}{\partial x} + \eta_i \frac{\partial}{\partial y}, \quad i = 1, 2, \dots, n \right\}$$

constitutes a Lie algebra if

$$\begin{aligned} [G_i, G_j] &= \left[\xi_i \frac{\partial}{\partial x} + \eta_i \frac{\partial}{\partial y}, \xi_j \frac{\partial}{\partial x} + \eta_j \frac{\partial}{\partial y} \right] \\ &= \left(\xi_i \frac{\partial \xi_j}{\partial x} + \eta_i \frac{\partial \xi_j}{\partial y} - \xi_j \frac{\partial \xi_i}{\partial x} - \eta_j \frac{\partial \xi_i}{\partial y} \right) \frac{\partial}{\partial x} \\ &\quad + \left(\xi_i \frac{\partial \eta_j}{\partial x} + \eta_i \frac{\partial \eta_j}{\partial y} - \xi_j \frac{\partial \eta_i}{\partial x} - \eta_j \frac{\partial \eta_i}{\partial y} \right) \frac{\partial}{\partial y} \\ &= C_{ijk} \left(\xi_k \frac{\partial}{\partial x} + \eta_k \frac{\partial}{\partial y} \right). \end{aligned}$$

How to deal with $f(x, y, y')$ and $f(x, y, y', y'')$

How do $y', y'', \dots$ transform?

$$(v) \quad a'''y^2 + c''y + d'' = 0. \quad (vii)$$

In (vi) and (vii) equate coefficients of independent powers of y to zero:

$$\begin{aligned} y^2 : a''' &= 0, \\ y^1 : c'' &= 0, \\ a'' &= 0, \\ y^0 : d'' &= 0, \\ b'' &= 2c'. \end{aligned}$$

$$\begin{aligned} a &= A_0 + A_1x, \\ b &= B_0 + B_1x + C_1x^2, \\ c &= C_0 + C_1x, \\ d &= D_0 + D_1x. \end{aligned}$$

$$\begin{aligned} \xi &= (A_0 + A_1x)y + B_0 + B_1x + C_1x^2, \\ \eta &= A_1y^2 + (C_0 + C_1x)y + D_0 + D_1x. \end{aligned}$$

Observations:

1. There are 8 generators which leave $y'' = 0$ invariant.
2. To obtain all of the symmetries it is necessary to be able to solve the original differential equation. $G^{(2)}N = 0$ is always linear in ξ and η .

For linear equations in general:

1. There are 8 symmetries.
2. The symmetries close under commutation and have the Lie algebra $\mathfrak{sl}(3, R)$.
3. Every linear equation may be transformed into every other linear equation by means of a point transformation.

[Algebras are invariant under point transformations]

4. The transformation is found by comparing the generators of the two equations. More specifically one looks for a subset.

For the free particle

$$\begin{aligned} G_1 &= \frac{\partial}{\partial x} & : B_0, \\ G_2 &= x \frac{\partial}{\partial x} & : B_1, \\ G_3 &= x^2 \frac{\partial}{\partial x} + xy \frac{\partial}{\partial y} & : C_1, \\ G_4 &= y \frac{\partial}{\partial x} & : A_0, \\ G_5 &= xy \frac{\partial}{\partial x} + y^2 \frac{\partial}{\partial y} & : A_1, \\ G_6 &= \frac{\partial}{\partial y} & : D_0, \\ G_7 &= x \frac{\partial}{\partial y} & : D_1, \\ G_8 &= y \frac{\partial}{\partial y} & : C_0. \end{aligned}$$

$$\eta' - y'\xi' = 0.$$

$$\frac{\partial\eta}{\partial x} + y'\frac{\partial\eta}{\partial y} - y'\frac{\partial\xi}{\partial x} - (y')^2\frac{\partial\xi}{\partial y} = 0.$$

$$(y')^2 : \quad \frac{\partial\xi}{\partial y} = 0 \quad (\text{i})$$

$$(y')^1 : \quad \frac{\partial\eta}{\partial y} - \frac{\partial\xi}{\partial x} = 0 \quad (\text{ii})$$

$$(y')^0 : \quad \frac{\partial\eta}{\partial x} = 0 \quad (\text{iii})$$

- (i) $\xi = a(x)$.
- (ii) $\eta = a'y + b(x)$.
- (iii) $a'' = 0, b' = 0$.

$$a = A_0 + A_1x, \quad b = B_0.$$

The generators

$$G_1 = \frac{\partial}{\partial x}, \quad A_0,$$

$$G_2 = \frac{\partial}{\partial y}, \quad B_0,$$

$$G_3 = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}, \quad A_1.$$

$$[G_1, G_2] = 0,$$

$$[G_1, G_3] = G_1,$$

$$[G_2, G_3] = G_2.$$

We wish to identify the complete symmetry group for the non-linear system (2.22). The generators of (2.22) given in the previous section should have commutation relations appropriate to one of the eight-parameter groups $SL(3, R)$, $SU(3)$ or $GL(2, C)$. The latter two being complex.

The commutation relations are given in Table 2. These relations are appropriate to the symmetry group $SL(3, R)$. Therefore the complete symmetry group for (2.22) is $SL(3, R)$. This will become apparent later.

Table 2

The entry in row i and column j is the Lie bracket $[G_i, G_j]$.

	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8
G_1	0	$-G_2$	$-G_3$	0	G_5	0	$G_7 - 2G_6$	G_3
G_2		0	0	G_2	$G_1 + G_4$	G_3	$3G_3 + G_8$	0
G_3			0	0	G_6	0	$2G_1 - G_4$	G_2
G_4				0	G_5	G_6	$2G_6$	$-(G_3 + G_8)$
G_5					0	0	0	$3G_6 - G_7$
G_6						0	G_5	$2G_4 - G_1$
G_7							0	$3G_4$
G_8								0

The entries below the main diagonal have been omitted because of the skewness of the Lie bracket. i.e. G_1, G_2, G_3 form a closed subgroup. For I_2 and I_3 the same result is found. (This gives 9 in all, but one is a linear combination of the others.) Can use this subalgebra to find a linearising transformation.

From the commutation table for the generators of

$$y'' + 3yy' + y^3 = 0$$

we see that

$$\begin{aligned} [G_1, G_2] &= -G_2, & [G_1, G_3] &= -G_3, \\ [G_2, G_3] &= 0. \end{aligned}$$

Define

$$\begin{aligned} [\tilde{G}_1, \tilde{G}_2] &= [G_2, G_3] = 0, \\ [\tilde{G}_1, \tilde{G}_3] &= [G_2, G_1] = G_2 = \tilde{G}_1, \\ [\tilde{G}_2, \tilde{G}_3] &= [G_3, G_1] = G_3 = \tilde{G}_2. \end{aligned}$$

i.e. we have the correct subalgebra.

In x, y coordinates we have

$$y'' + 3yy' + y^3 = 0,$$

$$\begin{aligned} \tilde{G}_1 &= y \frac{\partial}{\partial x} - y^3 \frac{\partial}{\partial y}, \\ \tilde{G}_2 &= xy \frac{\partial}{\partial x} - (xy^3 - y^2) \frac{\partial}{\partial y}, \\ \tilde{G}_3 &= \frac{1}{2}x^2y \frac{\partial}{\partial x} - \left(\frac{1}{2}x^2y^3 - xy^2 + y \right) \frac{\partial}{\partial y}. \end{aligned}$$

In X, Y coordinates we have

$$\begin{aligned} Y'' &= 0, \\ G_1 &= \frac{\partial}{\partial X}, \\ G_2 &= \frac{\partial}{\partial Y}, \\ G_3 &= X \frac{\partial}{\partial X} + Y \frac{\partial}{\partial Y}. \end{aligned}$$

$$\frac{\beta}{\alpha^2}Y'' + 3\frac{\beta^2}{\alpha}YY' + \beta^3Y^3 = 0,$$

or

$$Y'' + 3\alpha\beta YY' + (\alpha\beta)^2Y^3 = 0.$$

Any choice of $\alpha\beta$ will give a different equation which can also be linearised.

Let

$$\begin{aligned} y &= \frac{\omega'}{\omega}, \\ y' &= \frac{\omega''\omega - (\omega')^2}{\omega^2}, \\ y'' &= \frac{\omega'''\omega^2 - 3\omega''\omega'\omega + 2(\omega')^3}{\omega^3}. \end{aligned}$$

Thus

$$y'' + 3yy' + y^3 \longrightarrow \frac{\omega'''}{\omega} = 0,$$

so

$$\omega''' = 0, \quad \omega = A + Bx + Cx^2,$$

and

$$y = \frac{B + 2Cx}{A + Bx + Cx^2}.$$

This is a particular instance of a Riccati hierarchy. Recall the Riccati equation which is obtained from the general second-order linear differential equation

$$y'' + f(x)y' + g(x)y = 0.$$

Let

$$\frac{y'}{y} = u.$$

Then

$$\frac{y''}{y} - \frac{(y')^2}{y^2} = u', \quad \frac{y''}{y} = u' + u^2,$$

and hence

$$u' + u^2 + f(x)u + g(x) = 0,$$

the Riccati equation.

What nonlinear differential equations can have $SL(3, \mathbb{R})$ symmetry?

It must be possible to transform the differential equation to

$$Q'' = 0$$

by means of a point transformation. The reverse also applies.

Let

$$T = F(q, t), \quad Q = G(q, t).$$

Then $Q'' = 0$ becomes

$$\begin{aligned} & \ddot{q} + \dot{q}^3 J^{-1} \left(\frac{\partial F}{\partial q} \frac{\partial^2 G}{\partial q^2} - \frac{\partial^2 F}{\partial q^2} \frac{\partial G}{\partial q} \right) \\ & + \dot{q}^2 J^{-1} \left(\frac{\partial F}{\partial t} \frac{\partial^2 G}{\partial q^2} + 2 \frac{\partial F}{\partial q} \frac{\partial^2 G}{\partial q \partial t} - 2 \frac{\partial^2 F}{\partial q \partial t} \frac{\partial G}{\partial q} - \frac{\partial^2 F}{\partial q^2} \frac{\partial G}{\partial t} \right) \\ & + \dot{q} J^{-1} \left(2 \frac{\partial F}{\partial t} \frac{\partial^2 G}{\partial q \partial t} + \frac{\partial F}{\partial q} \frac{\partial^2 G}{\partial t^2} - \frac{\partial^2 F}{\partial t^2} \frac{\partial G}{\partial q} - 2 \frac{\partial^2 F}{\partial q \partial t} \frac{\partial G}{\partial t} \right) \\ & + J^{-1} \left(\frac{\partial F}{\partial t} \frac{\partial^2 G}{\partial t^2} - \frac{\partial^2 F}{\partial t^2} \frac{\partial G}{\partial t} \right) = 0, \end{aligned}$$

where

$$J = \frac{\partial(F, G)}{\partial(t, q)} \neq 0.$$

The general third-order constant-coefficient equation

$$\omega''' + a\omega'' + b\omega' + c\omega = 0$$

has a solution. Let

$$\frac{\omega'}{\omega} = y.$$

Then

$$\frac{\omega'}{\omega} = y, \quad \frac{\omega''}{\omega} = y' + y^2,$$

and

$$\begin{aligned} \frac{\omega'''}{\omega} &= y'' + 2yy' + y(y' + y^2) \\ &= y'' + 3yy' + y^3. \end{aligned}$$

Thus

$$y'' + 3yy' + y^3 + a(y' + y^2) + by + c = 0$$

can be linearised.

Aside: $\omega''' = 0$ has seven symmetries.

What second-order ODEs can be linearised?

$$y'' = f(x, y, y').$$

(Transparency.)

Classify by extra information

1. Existence of one symmetry Transform the differential equation to standard form by making the symmetry, say, $\partial/\partial x$. Then the differential equation is

$$y'' = f(y, y').$$

2. Existence of two symmetries The possibilities include

$$G_1 = x \frac{\partial}{\partial x}, \quad G_2 = \frac{\partial}{\partial x}, \quad [G_1, G_2] = -G_2,$$

and

$$G_1 = \frac{\partial}{\partial x}, \quad G_2 = \frac{\partial}{\partial y}, \quad [G_1, G_2] = 0,$$

with $G_2 \neq f(x, y)G_1$.

Two commuting symmetries

$$[G_1, G_2] = 0.$$

Suppose we choose coordinates such that

$$G_1 = \frac{\partial}{\partial y}, \quad G_2 = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}.$$

Then

$$[G_1, G_2] = \frac{\partial \xi}{\partial y} \frac{\partial}{\partial x} + \frac{\partial \eta}{\partial y} \frac{\partial}{\partial y} = 0,$$

so

$$\frac{\partial \xi}{\partial y} = 0, \quad \frac{\partial \eta}{\partial y} = 0.$$

Observe that Type II is already linear and the solution is reduced to two quadratures.

Two non-commuting symmetries

$$[G_1, G_2] = G_1.$$

Let the coordinates be such that

$$G_1 = \frac{\partial}{\partial y}, \quad G_2 = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}.$$

Then $[G_1, G_2] = G_1$ becomes

$$\frac{\partial \xi}{\partial y} \frac{\partial}{\partial x} + \frac{\partial \eta}{\partial y} \frac{\partial}{\partial y} = \frac{\partial}{\partial y},$$

so

$$\frac{\partial \xi}{\partial y} = 0, \quad \frac{\partial \eta}{\partial y} = 1.$$

Thus

$$\xi = a(x), \quad \eta = b(x) + y,$$

and

$$G_2 = a(x) \frac{\partial}{\partial x} + (b(x) + y) \frac{\partial}{\partial y}.$$

The two forms written for G_2 are

$$\begin{aligned} G_2 &= y \frac{\partial}{\partial y} = y G_1, \\ G_2 &= x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \neq f(x, y) G_1. \end{aligned}$$

Note that Type IV is linear. Type I,

$$y'' = f(y'),$$

and Type III,

$$y'' = \frac{1}{x} f(y'),$$

are the ones whose linearisation is still an open question.

Proposition

In order that a second-order ODE have the symmetry algebra $\mathfrak{sl}(3, \mathbb{R})$, and hence be linearisable, it is necessary and sufficient that it have the three-element subalgebra $\mathcal{X}$ of generators G_1, G_2, G_3 satisfying

$$[G_1, G_2] = 0, \quad [G_2, G_3] = 0, \quad [G_3, G_1] = G_2.$$

Proof

For

$$y'' = f(x, y, y'),$$

a generator of symmetry is

$$G = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}.$$

Thus

$$G_3 = A \frac{\partial}{\partial x} + (B - x) \frac{\partial}{\partial y}.$$

Take

$$G_3 = -x \frac{\partial}{\partial y}.$$

The equation

$$y'' = f(x, y, y')$$

is invariant under these three generators. From

$$G_1^{(2)}[y'' - f(x, y, y')] = 0$$

we obtain

$$\frac{\partial f}{\partial x} = 0.$$

From

$$G_2^{(2)}[y'' - f] = 0$$

we obtain

$$\frac{\partial f}{\partial y} = 0.$$

Finally,

$$G_3^{(2)}[y'' - f] = 0$$

gives

$$\left(-x \frac{\partial}{\partial y} - \frac{\partial}{\partial y'}\right)[y'' - f] = 0,$$

so

$$x \frac{\partial f}{\partial y} + \frac{\partial f}{\partial y'} = 0.$$

Therefore f is constant and the differential equation is

$$y'' = \text{constant},$$

which is linear. Hence the symmetry of the original differential equation is $\mathfrak{sl}(3, \mathbb{R})$. We have proven sufficiency, i.e. the possession of G_1, G_2, G_3 such that

$$[G_1, G_2] = 0, \quad [G_2, G_3] = 0, \quad [G_3, G_1] = G_2.$$

Necessity follows from $\mathcal{X}$ being a subalgebra of $\mathfrak{sl}(3, \mathbb{R})$.

and from (iii),

$$\begin{aligned}\xi_y &= \frac{1}{3d} (2\eta_{xy} - \xi_{xx} - c\xi_x) \\ &= \frac{1}{3d} \left[2\eta_{xy} - \frac{1}{2d} (\eta_{xxx} + d\eta_{xy} - c\eta_{xx}) - \frac{c}{2d} (\eta_{xx} + d\eta_y - c\eta_x) \right].\end{aligned}$$

Eliminate η . From (i),

$$\eta_y = \frac{1}{2}(\xi_x - \xi_{yy}), \quad (\text{v})$$

and from (ii),

$$\begin{aligned}\eta_x &= \frac{1}{3}(\eta_{yy} - 2c\xi_y - 2\xi_{xy}) \\ &= -\frac{1}{6}(\xi_{yyy} + 3\xi_{xy} + 4c\xi_y),\end{aligned} \quad (\text{vi})$$

using (v). Substitution of (v) and (vi) in (iii) and (iv) gives

$$\xi_{xyy} + 3d\xi_y + c\xi_x = 0, \quad (\text{vii})$$

and

$$\xi_{xyyy} - c\xi_{yyy} + 3\xi_{xxy} + 3d\xi_{yy} + c\xi_{xy} - 4c^2\xi_y + 9d\xi_x = 0. \quad (\text{viii})$$

The consistency condition between (v) and (vi) is

$$\frac{\partial}{\partial x}(\eta_y) = \frac{\partial}{\partial y}(\eta_x),$$

which gives

$$\xi_{yyy} + 4c\xi_{yy} + 3\xi_{xx} = 0. \quad (\text{ix})$$

Using the y derivative of (vii), equation (viii) simplifies to

$$-c\xi_{yyy} + 3\xi_{xxy} - 4c^2\xi_y + 9d\xi_x = 0. \quad (\text{x})$$

Also,

$$\frac{\partial^2}{\partial y^2}(\text{vii}) - \frac{\partial}{\partial x}(\text{ix})$$

gives

$$\xi_{xxx} + c\xi_{xyy} - d\xi_{yyy} = 0. \quad (\text{xi})$$

Use (vii), (x), and (xi), which can be shown to be equivalent to (vii), (viii), and (ix), except when both c and d are zero.

For the case $c = d = 0$,

$$\xi_{xyy} = 0, \quad (\text{vii})$$

and

$$\xi_{xyyy} + 3\xi_{xxy} = 0. \quad (\text{viii})$$

Equation (ix) becomes

$$\xi_{yyyy} + 3\xi_{xx} = 0. \quad (\text{ix})$$

From (vii),

$$\xi = F(y) + yG(x) + H(x).$$

Equation (viii) gives

$$G'' = 0.$$

In (ix),

$$F^{(4)} + 3yG'' + H'' = 0.$$

Therefore

$$\begin{aligned} G &= G_0 + G_1x, \\ H &= H_0 + H_1x + \frac{H_2}{2}x^2, \\ F &= F_1 + F_2y + F_3y^2 + F_4y^3 - \frac{H_2}{24}y^4. \end{aligned}$$

Thus

$$\begin{aligned} \xi &= F_1 + F_2y + F_3y^2 + F_4y^3 - \frac{H_2}{24}y^4 + G_0y + G_1xy \\ &\quad + H_0 + H_1x + \frac{H_2}{2}x^2 \\ &= (F_1 + H_0) + (F_2 + G_0)y + F_3y^2 + F_4y^3 + G_1xy + H_1x \\ &\quad + H_2\left(\frac{1}{2}x^2 - \frac{1}{24}y^4\right). \end{aligned}$$

The eighth generator comes from

$$\eta_x = 0, \quad \eta_y = 0$$

when $\xi = 0$, i.e. η is constant. This gives

$$G_1 = \frac{\partial}{\partial y}.$$

For

$$y'' = (y')^3,$$

find the linearising transformation.

Note that we have

Observe that (xi) is homogeneous in order. Hence it can be solved by factorisation. Suppose it factors as

$$\left(\frac{\partial}{\partial x} + \alpha \frac{\partial}{\partial y}\right) \left(\frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y}\right) \left(\frac{\partial}{\partial x} + \gamma \frac{\partial}{\partial y}\right) \xi = 0.$$

Corresponding to each factor there is an equation for a characteristic. However, one must be careful of repeated roots. Observe that

$$\alpha\beta\gamma = -d, \quad \alpha\beta + \beta\gamma + \gamma\alpha = c, \quad \alpha + \beta + \gamma = 0.$$

A triple root gives $c = d = 0$, which has already been treated.

For two roots α and -2α ,

$$c = -3\alpha^2, \quad d = -2\alpha^3,$$

and

$$\left(\frac{\partial}{\partial x} + \alpha \frac{\partial}{\partial y}\right)^2 \left(\frac{\partial}{\partial x} - 2\alpha \frac{\partial}{\partial y}\right) \xi = 0.$$

The characteristic equations give

$$u_1 = \alpha x - y, \quad u_2 = 2\alpha x + y.$$

The solution is

$$\xi = f(\alpha x - y) + g(2\alpha x + y) + (2\alpha x + y)h(\alpha x - y).$$

Substitution in (xi) and (x) gives

$$\begin{aligned} & \alpha f''' + 2\alpha g''' + 2\alpha h'' + \alpha h'' - \alpha h''' + \alpha(2\alpha x + y)h''' \\ & - 6\alpha^3[-f' + g' + h - (2\alpha x + y)h'] \\ & - 3\alpha^2[\alpha f'' + 2\alpha g'' + 2\alpha h + \alpha(2\alpha x + y)h''] = 0. \end{aligned}$$

The separate coefficients give

$$2\alpha g''' - 12\alpha^3 g' = 0,$$

$$(2\alpha x + y)(h''' + 3\alpha^3 h') = 0,$$

and

$$\alpha f''' + 2\alpha h'' + 3\alpha^3 f' = 0.$$

The remaining determining equation is

$$\begin{aligned} 0 = & 3\alpha^2[-f''' + g''' + 3h'' - (2\alpha x + y)h'''] \\ & + 3[-\alpha^2 f''' + 4\alpha^2 g''' - 3\alpha^2 h'' - \alpha^2(2\alpha x + y)h'''] \\ & + 36\alpha^4[-f' + g' + h - (2\alpha x + y)h'] \\ & - 18\alpha^3[\alpha f'' + 2\alpha g'' + 2\alpha h' + \alpha(2\alpha x + y)h'']. \end{aligned} \tag{ix}$$

The correct equations are

$$g''' = 0, \quad h''' + 9\alpha^2 h' = 0, \quad f''' + 9\alpha^2 f' = 0.$$

The solutions written in the notes are

$$\begin{aligned} g &= G_0 + G_1(2\alpha x + y) + G_2(2\alpha x + y)^2, \\ h &= [H_0 + H_1 \sin 3\alpha(\alpha x - y) + H_2 \cos 3\alpha(\alpha x - y)](2\alpha x + y), \\ f &= f_0 + F_1(\alpha x - y) + F_2(\alpha x - y)^2. \end{aligned}$$

There are seven independent solutions and so seven generators. The eighth comes from $\xi = 0$, giving η constant.

For three distinct roots,

$$-2\alpha, \quad \alpha + \delta, \quad \alpha - \delta,$$

where $\delta \neq 0, \pm 3\alpha$ to avoid repeated roots, equation (xi) becomes

$$\left(\frac{\partial}{\partial x} - 2\alpha \frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} + (\alpha + \delta) \frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} + (\alpha - \delta) \frac{\partial}{\partial y} \right) \xi = 0,$$

where

$$c = \delta^2 - 3\alpha^2, \quad d = 2\alpha(\alpha^2 - \delta^2).$$

The characteristics give

$$u_1 = 2\alpha x + y, \quad u_2 = (\alpha + \delta)x - y, \quad u_3 = (\alpha - \delta)x - y,$$

and

$$\xi = f(2\alpha x + y) + g((\alpha + \delta)x - y) + h((\alpha - \delta)x - y).$$

Substitution into (vii) and (xi) again gives seven generators, which become eight with η constant. Hence

$$y'' = (y')^3 + cy' + d$$

is always linearisable.

From Type 2, consider

$$y'' = (y')^2 + d.$$

Invariance under

$$G = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}$$

gives

$$Y' = \frac{-2\omega e^{2\omega x}}{(y' - \omega)e^{-y+\omega x}},$$

$$Y'' = \frac{-4\omega^2 e^{2\omega x}(y' - \omega)e^{-y+\omega x} + 2\omega e^{2\omega x} [y'' - (y' - \omega)^2] e^{-y+\omega x}}{[(y' - \omega)e^{-y+\omega x}]^3}.$$

The numerator is

$$\begin{aligned} 2\omega e^{-y+3\omega x} [-2\omega y' + 2\omega^2 + y'' - (y')^2 + 2\omega y' - \omega^2] \\ = 2\omega e^{-y+3\omega x} [y'' - (y')^2 + \omega^2], \end{aligned}$$

as required.

The solution of $Y'' = 0$ is

$$Y = A + BX.$$

The solution of $y'' = (y')^2 - \omega^2$ is therefore

$$\begin{aligned} -e^{2\omega x} &= A + Be^{-y+\omega x}, \\ e^{-y} &= Ce^{\omega x} + De^{-\omega x}, \\ y &= -\log(Ce^{\omega x} + De^{-\omega x}). \end{aligned}$$

Check:

$$\begin{aligned} \frac{y''}{(y')^2 - \omega^2} &= 1, \\ \frac{d((y')^2)}{(y')^2 - \omega^2} &= 2 dy, \\ \log((y')^2 - \omega^2) &= 2y + k, \\ (y')^2 &= \omega^2 + me^{2y}. \end{aligned}$$

Thus

$$I = \int \frac{dy}{(\omega^2 + me^{2y})^{1/2}} = \int dx.$$

Let

$$e^{-y} = u, \quad -e^{-y} dy = du, \quad dy = -\frac{du}{u}.$$

Then

$$\begin{aligned} I &= \int \frac{-u^{-1} du}{(\omega^2 + mu^{-2})^{1/2}} = - \int \frac{du}{(\omega^2 u^2 + m)^{1/2}}. \\ I &= -\frac{1}{\omega} \operatorname{arsinh}\left(\frac{\omega u}{m^{1/2}}\right) = x + c. \end{aligned}$$

Therefore

Proof

We cannot have both

$$H = a_i G_i \quad \text{and} \quad X_i = b_{ij} G_j$$

for suitable functions a_i and b_{ij} . For, from (2),

$$\begin{aligned} [H, G_j] &= [a_i G_i, G_j] \\ &= a_i [G_i, G_j] - (G_j a_i) G_i \\ &= -(G_j a_i) G_i \\ &= 0, \end{aligned}$$

and, from (4),

$$\begin{aligned} [G_i, X_j] &= [G_i, b_{jk} G_k] \\ &= [G_i, G_k] b_{jk} + (G_i b_{jk}) G_k \\ &= (G_i b_{jk}) G_k \\ &= 0. \end{aligned}$$

Then (5) is

$$\begin{aligned} [H, X_j] &= [a_i G_i, b_{jk} G_k] \\ &= a_i [G_i, G_k] b_{jk} + a_i (G_i b_{jk}) G_k - b_{jk} (G_k a_i) G_i \\ &= 0, \end{aligned}$$

which is a contradiction.

We assume that $H \neq a_i G_i$. Then there exist coordinates x, u_i such that

$$H = \frac{\partial}{\partial x}, \quad G_j = C_{ij} \frac{\partial}{\partial u_i}.$$

Since

$$[H, G_j] = \frac{\partial C_{ij}}{\partial x} \frac{\partial}{\partial u_i} = 0,$$

it follows that

$$\frac{\partial C_{ij}}{\partial x} = 0,$$

so $C_{ij} = C_{ij}(\mathbf{u})$. Since the G_i are linearly independent, the matrix $[C_{ij}]$ has rank n , and so its inverse exists, perhaps restricted to a subset of $\mathbb{R}^n$.

If we change variables from u_i to v_i so that

$$v_i = d_i(\mathbf{u}),$$

then

$$\begin{aligned}
G_i &= C_{ik} \frac{\partial}{\partial u_k} \\
&= C_{ik} \frac{\partial d_j}{\partial u_k} \frac{\partial}{\partial v_j} \\
&= \frac{\partial}{\partial v_i},
\end{aligned}$$

provided

$$C_{ik} \frac{\partial d_j}{\partial u_k} = \delta_{ij},$$

i.e. the Jacobian of the transformation is the inverse of $[C_{ij}]$. Thus the transformation is not degenerate. Can functions $d_j(\mathbf{u})$ exist with the property

$$\frac{\partial d_i}{\partial u_j} = (C^{-1})_{ij}, \quad C = [C_{ij}]?$$

The requirement of consistency, if the integration is to be carried out, is

$$\frac{\partial}{\partial u_k} \frac{\partial d_i}{\partial u_j} = \frac{\partial}{\partial u_j} \frac{\partial d_i}{\partial u_k}.$$

Now

$$\frac{\partial C^{-1}}{\partial u_i} = -C^{-1} \frac{\partial C}{\partial u_i} C^{-1}.$$

The differentiability of C^{-1} depends upon the differentiability of C . Since we need C''' in $G^{(2)}$, C must be twice differentiable almost everywhere, and so d_i is three times differentiable and the mixed derivatives are equal. Thus a transformation exists which casts G_i in the form

$$G_i = \frac{\partial}{\partial u_i}.$$

Had

$$X_i = b_{ij} \frac{\partial}{\partial u_j},$$

then

$$\begin{aligned}
[G_i, X_j] &= \left[\frac{\partial}{\partial u_i}, b_{jk} \frac{\partial}{\partial u_k} \right] \\
&= \frac{\partial b_{jk}}{\partial u_i} \frac{\partial}{\partial u_k} = 0.
\end{aligned}$$

Hence

$$\frac{\partial b_{jk}}{\partial u_i} = 0, \quad b_{jk} = B_{jk}(x).$$

From (5),

$$\eta' = \frac{\partial \eta}{\partial x} + u' \frac{\partial \eta}{\partial u} + v' \frac{\partial \eta}{\partial v},$$

and

$$\begin{aligned} \eta'' = & \frac{\partial^2 \eta}{\partial x^2} + 2u' \frac{\partial^2 \eta}{\partial u \partial x} + 2v' \frac{\partial^2 \eta}{\partial v \partial x} + (u')^2 \frac{\partial^2 \eta}{\partial u^2} \\ & + 2u'v' \frac{\partial^2 \eta}{\partial u \partial v} + (v')^2 \frac{\partial^2 \eta}{\partial v^2} + u'' \frac{\partial \eta}{\partial u} + v'' \frac{\partial \eta}{\partial v}. \end{aligned}$$

The action of $G^{(2)}$ on the two equations, followed by separation by powers of u' and v' , leads to a system of fifteen independent partial differential equations for τ , ξ , and η , all linear. The solutions give

$$\begin{aligned} G_1 &= x \frac{\partial}{\partial x} + 2u \frac{\partial}{\partial u} + 2v \frac{\partial}{\partial v}, \\ G_2 &= \frac{\partial}{\partial x}, \\ G_3 &= \frac{1}{v} \frac{\partial}{\partial u} + \frac{1}{u} \frac{\partial}{\partial v}, \\ G_4 &= x \frac{\partial}{\partial x}, \\ G_5 &= x^2 \frac{\partial}{\partial x} + 2xu \frac{\partial}{\partial u} + 2xv \frac{\partial}{\partial v}, \\ G_6 &= \frac{x}{v} \frac{\partial}{\partial u} + \frac{x}{u} \frac{\partial}{\partial v}, \\ G_7 &= xuv \frac{\partial}{\partial x} + 2u^2v \frac{\partial}{\partial u} + 2uv^2 \frac{\partial}{\partial v}, \\ G_8 &= uv \frac{\partial}{\partial x}, \\ G_9 &= x \frac{\partial}{\partial x} + 2u \frac{\partial}{\partial u} - 2v \frac{\partial}{\partial v}, \\ G_{10} &= v \frac{\partial}{\partial u} - \frac{v^2}{u} \frac{\partial}{\partial v}, \\ G_{11} &= x^2 \frac{\partial}{\partial x} + 2ux \frac{\partial}{\partial u} - 2vx \frac{\partial}{\partial v}, \\ G_{12} &= vx \frac{\partial}{\partial u} - \frac{v^2x}{u} \frac{\partial}{\partial v}, \\ G_{13} &= \frac{xv}{u} \frac{\partial}{\partial x} + \frac{2u^2}{v} \frac{\partial}{\partial u} - 2u \frac{\partial}{\partial v}, \\ G_{14} &= \frac{u}{v} \frac{\partial}{\partial x}, \\ G_{15} &= u \left(v^2 + \frac{1}{v^2} \right) \frac{\partial}{\partial u} - v \left(u^2 - \frac{1}{v^2} \right) \frac{\partial}{\partial v}. \end{aligned}$$

The system is obviously linearisable, as the fifteen-element algebra is $\mathfrak{sl}(4, \mathbb{R})$, the algebra of the two-dimensional free particle with equation of motion

$$\mathbf{r}'' = 0, \quad \mathbf{r} \in \mathbb{R}^2.$$

Note that $\mathfrak{sl}(4, \mathbb{R})$ is not a prerequisite for linearisation. This is because the system

$$\mathbf{r}'' = A\mathbf{r}, \quad \mathbf{r} \in \mathbb{R}^2,$$

need not possess $\mathfrak{sl}(4, \mathbb{R})$ symmetry.

From the symmetries, one looks to see if the algebra $\mathcal{N}$ is present. Recall that $\mathcal{N}$ has H, G_i, X_i such that

$$[H, G_i] = 0, \quad [G_i, G_j] = 0,$$

and

$$[G_i, X_j] = 0, \quad [H, X_i] = G_i.$$

In standard form,

$$H = \frac{\partial}{\partial x}, \quad G_i = \frac{\partial}{\partial u_i}, \quad X_i = x \frac{\partial}{\partial u_i}.$$

Noting that

$$[G_2, G_3] = 0, \quad [G_2, G_6] = G_3, \quad [G_3, G_6] = 0,$$

and

$$[G_2, G_{10}] = 0, \quad [G_2, G_{12}] = G_{10}, \quad [G_{10}, G_{12}] = 0,$$

we check to see if this is the algebra $\mathcal{N}$:

$$\begin{aligned} [G_3, G_{10}] &= \left[\frac{1}{v} \frac{\partial}{\partial u} + \frac{1}{u} \frac{\partial}{\partial v}, v \frac{\partial}{\partial u} - \frac{v^2}{u} \frac{\partial}{\partial v} \right] \\ &= \left(\frac{1}{u} - \frac{1}{u} \right) \frac{\partial}{\partial u} + \left(\frac{v}{u^2} - \frac{2v}{u^2} + \frac{v}{u^2} \right) \frac{\partial}{\partial v} \\ &= 0. \end{aligned}$$

Likewise,

$$[G_6, G_{10}] = 0, \quad [G_6, G_{12}] = 0, \quad [G_3, G_{12}] = 0.$$

Thus we do have $\mathcal{N}$. Identify

$$\begin{aligned} H &= \frac{\partial}{\partial x}, \\ G_1 &= \frac{1}{v} \frac{\partial}{\partial u} + \frac{1}{u} \frac{\partial}{\partial v}, \\ G_2 &= v \frac{\partial}{\partial u} - \frac{v^2}{u} \frac{\partial}{\partial v}, \\ X_1 &= x G_1, \\ X_2 &= x G_2. \end{aligned}$$

Seek a transformation from x, u, v to X, U, V such that

$$\begin{aligned} H &= \frac{\partial}{\partial X}, \quad G_1 = \frac{\partial}{\partial U}, \quad G_2 = \frac{\partial}{\partial V}, \\ X_1 &= X \frac{\partial}{\partial U}, \quad X_2 = X \frac{\partial}{\partial V}. \end{aligned}$$

Let

$$X = x, \quad U = F(u, v), \quad V = G(u, v),$$

having already realised that x is an appropriate variable.

The derivatives transform according to

$$\frac{\partial}{\partial u} \longrightarrow \frac{\partial F}{\partial u} \frac{\partial}{\partial U} + \frac{\partial G}{\partial u} \frac{\partial}{\partial V},$$

and

$$\frac{\partial}{\partial v} \longrightarrow \frac{\partial F}{\partial v} \frac{\partial}{\partial U} + \frac{\partial G}{\partial v} \frac{\partial}{\partial V}.$$

Thus G_1 gives

$$\frac{1}{v} \frac{\partial F}{\partial u} + \frac{1}{u} \frac{\partial F}{\partial v} = 1, \quad (\text{i})$$

and

$$\frac{1}{v} \frac{\partial G}{\partial u} + \frac{1}{u} \frac{\partial G}{\partial v} = 0. \quad (\text{ii})$$

From G_2 we obtain

$$v \frac{\partial F}{\partial u} - \frac{v^2}{u} \frac{\partial F}{\partial v} = 0, \quad (\text{iii})$$

and

$$v \frac{\partial G}{\partial u} - \frac{v^2}{u} \frac{\partial G}{\partial v} = 1. \quad (\text{iv})$$

For (ii),

$$\frac{du}{1/v} = \frac{dv}{1/u}, \quad w_1 = \frac{u}{v},$$

so

$$G = g\left(\frac{u}{v}\right).$$

In (iv),

$$v \frac{1}{v} g' - \frac{v^2}{u} \left(-\frac{u}{v^2}\right) g' = 1,$$

so

$$g' = \frac{1}{2},$$

and

$$G = \frac{1}{2} \frac{u}{v},$$

$$U = A_0 + A_1x,$$

$$V = B_0 + B_1x,$$

giving

$$u^2 = 4(A_0 + A_1x)(B_0 + B_1x),$$

$$v^2 = \frac{A_0 + A_1x}{B_0 + B_1x},$$

which is the general solution since there are four arbitrary constants.

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